



RUSSIAN-AFRICAN FORUM OF YOUNG SCIENTISTS: "FUTURE ENGINEERS OF THE WORLD –
THE FOUNDATION OF SUSTAINABLE DEVELOPMENT"

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PROJECT TITLE

SECURING THE DIGITAL MINE: A METAHEURISTIC OPTIMIZED DEEP LEARNING
FRAMEWORK FOR INTRUSION DETECTION IN IOT ENABLED MINERAL
RESOURCE OPERATIONS

Nomination: Track 3, "Smart Subsoil": Digital Transformation and Automation in the Mineral
Resources Complex

Abstract. African mining operations are digitalizing rapidly through IoT sensor networks, SCADA systems, and cloud connected digital twins, yet cybersecurity for these operational technology environments lags behind conventional IT networks (Alanazi, Mahmood, & Chowdhury, 2022). This work proposes adapting a Binary Whale Optimization Algorithm combined with a CNN-LSTM classifier from generic network intrusion detection toward the distinct traffic patterns of mining IoT and SCADA infrastructure. A three-phase adaptation roadmap is outlined, required data partnerships are identified, and the proposal is mapped to relevant UN Sustainable Development Goals.

Keywords: intrusion detection; metaheuristic optimization; Whale Optimization Algorithm; CNN-LSTM; Industrial IoT; SCADA; mining digitalization; cybersecurity; Africa

Introduction. Mining and mineral processing operations across Africa, including Ghana, are undergoing rapid digital transformation: IoT sensors monitor equipment and ore bodies, AI assisted predictive maintenance is deployed at sites such as Gold Fields' Tarkwa mine, and digital twins support ecological and operational risk management (Nigerian Mineral Exchange, 2025; IT-Online, 2026). This same connectivity expands the attack surface of facilities that were historically isolated from external networks. Industry reporting confirms that digitalization has already widened mining sector exposure to cyberattacks capable of manipulating programmable logic controllers, halting



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production, or endangering worker safety (African Mining Market, 2024). Meanwhile, most intrusion detection research and benchmark datasets target conventional IT and cloud traffic rather than the protocols, polling intervals, and sensor-driven traffic signatures characteristic of operational technology (OT) and Industrial IoT (IIoT) environments (Kheddar, Himeur, & Awad, 2023).

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For African mining operators adopting digital monitoring under constrained infrastructure and connectivity conditions (IT-Online, 2026), this gap is consequential. This project asks how an intrusion detection approach already validated on conventional network traffic can be systematically adapted to the operational realities of digitalizing African mining infrastructure.

Methods. The starting point is a Binary Whale Optimization Algorithm (BWOA) combined with a CNN-LSTM classifier, validated on the NSL-KDD benchmark for conventional network intrusion detection. The Whale Optimization Algorithm, introduced by Mirjalili and Lewis (2016) as a bio-inspired metaheuristic mimicking humpback whale bubble-net hunting, has since been applied widely to intrusion detection feature selection. Hybrid WOA approaches combined with CNN, LSTM, or ensemble classifiers have reported accuracy above 96 percent on NSL-KDD and related benchmarks (Krishnaveni et al., 2025; Anand & Arul, 2024). The CNN-LSTM architecture captures both spatial packet-level patterns and temporal sequence dynamics, and has been applied successfully to SCADA and IIoT intrusion detection contexts (Almomani, Akour, & Habeb, 2025). A systematic review of deep transfer learning for industrial control network intrusion detection (Kheddar, Himeur, & Awad, 2023) identifies the structural differences between IT and OT traffic, including deterministic polling cycles, industrial protocol fields, and narrower attack-class diversity, that any adaptation must address. Building on this analysis, we propose a methodology for retraining and validating the BWOA plus CNN-LSTM pipeline on mining-relevant OT and IIoT traffic, incorporating a plan for collaborative data collection with mining operators. Novelty lies not in a new algorithm but in a concrete, literature-grounded adaptation pathway suited to under-resourced African mining contexts where labeled OT attack data remains scarce.

Results. A three-phase adaptation framework is proposed. Phase 1 (Data Partnership): instrumented OT traffic capture at pilot mining sites or testbeds in collaboration with industry and academic partners, using a two-instance AWS EC2 and CICFlowMeter architecture to generate labeled attack and benign traffic for SCADA protocols (Modbus, DNP3, OPC-UA) and IoT sensor streams. Phase 2 (Model Adaptation): retraining the BWOA feature selection stage and the CNN-LSTM classifier on OT-specific feature sets, with validation against established IIoT intrusion benchmarks including SWaT and BATADAL, following methodological precedents in recent WOA-based and CNN-LSTM-based intrusion detection studies (Krishnaveni et al., 2025; Almomani, Akour, & Habeb, 2025). Phase 3 (Deployment Readiness): evaluating detection latency and computational footprint against edge-deployment constraints at remote mining sites with limited connectivity and power infrastructure (IT-Online, 2026), targeting sub-100 ms detection latency on Raspberry Pi-class hardware.

Discussion. This proposal aligns with three UN Sustainable Development Goals. SDG 9 (Industry, Innovation and Infrastructure): strengthening resilience of digitalizing industrial infrastructure. SDG 8 (Decent Work and Economic Growth): protecting operational continuity and worker safety in mining operations, given documented risks from OT compromise (African Mining Market, 2024). SDG 17 (Partnerships for the Goals): the proposed Russian-African data sharing and research pathway directly embodies cross-continental scientific cooperation. Economically, undetected ICS intrusions risk production halts and costly equipment damage. Environmentally, compromised control systems can trigger unmonitored ecological impact from mining and processing operations. Socially, securing these systems protects the livelihoods and safety of workers and communities dependent on African mining economies, particularly as digitalization continues to outpace cybersecurity investment in the sector (IT-Online, 2026). The team brings direct validated experience in metaheuristic-optimized deep learning for intrusion detection, grounding this as a concrete proposal rather than a speculative one.

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Conclusion. As African and Russian mining operations digitalize, cybersecurity for OT and IIoT environments must keep pace with urgency already applied to conventional IT systems. The target audience encompasses mining operators, industrial cybersecurity teams, and policymakers shaping digital infrastructure strategy across both continents. The framework is designed for practical adaptation to African mining conditions: edge-first deployment, offline-capable detection, and human-readable attack class outputs for non-specialist operators. A Phase 3 localization component provides for training local cybersecurity personnel at partner mining sites, ensuring the solution builds African technical capacity rather than merely transferring a foreign tool. Practical significance lies in offering a concrete, resource-conscious starting point for securing the digital mine, treating cybersecurity as a core component of digitalization strategy rather than its afterthought.

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